

# Operations Research

Convex optimization

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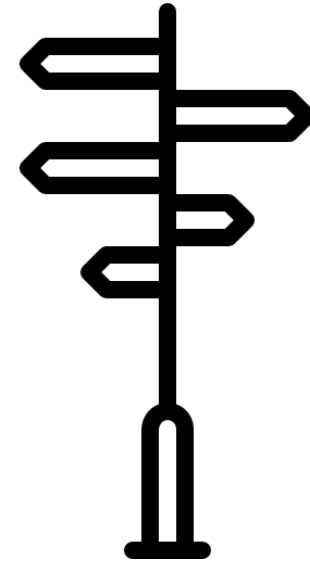
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# Course outline

- **Linear Optimization**
  - Modeling by linear programming and graphical solution
  - Solving linear programs, convexity
  - Simplex algorithm, 2-phase simplex
  - Simplex algorithm in matrix notation
  - Sensitivity analysis, reoptimization
  - Duality theory, zero-sum games
- **Integer Linear Optimization**
  - Modeling integer programming problems (IPs)
  - Solving IPs: branch & bound, cutting planes
  - Advanced solution methods: column generation, approximation algorithms
  - Fast solvable integer problems: total unimodularity, matroids
  - Network flow problems
  - Traveling salesperson problem
- **Nonlinear Optimization**
  - Optimization of multidimensional functions
  - **Convex optimization**



# Convex optimization (Recap)

## Definition (Convex optimization):

Let  $X \subset \mathbb{R}^n$  be a convex set and  $f: X \rightarrow \mathbb{R}$  a convex function.

Compute the minimum of the function  $f$  over the feasible set  $X$ :  $\min_{x \in X} f(x)$ .

All local solutions of convex optimization problems are also global.

## Special cases

- if  $X = \mathbb{R}^n$  we talk about unconstrained optimization
- linear optimization is a special case of convex optimization
- $\exp x, -\log x, x \log x$
- norms and projections

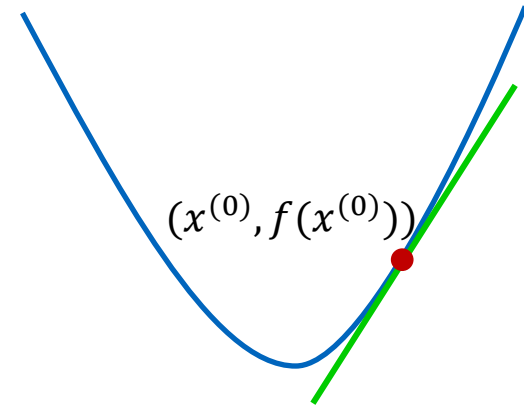
# Gradient descent method

Task:  $\min f(x), \text{ s.t. } x \in X = \mathbb{R}^n$

## Derivation via 1st order Taylor approximation

(i.e., via a linear function)

$$f(x) \approx f(x^{(0)}) + \langle \nabla f(x^{(0)}), x - x^{(0)} \rangle$$



This would follow the gradient arbitrarily far. However, the approximation is only good around  $x^{(0)}$ . Therefore, we add a penalty term and use the following formula, starting from  $k = 0$ .

$$\begin{aligned} x^{(k+1)} &= \operatorname{argmin}_x f(x^{(k)}) + \langle \nabla f(x^{(k)}), x - x^{(k)} \rangle + \frac{1}{2\eta} \|x - x^{(k)}\|^2 \\ &= \operatorname{argmin}_x \langle \nabla f(x^{(k)}), x \rangle + \frac{1}{2\eta} \|x - x^{(k)}\|^2 \end{aligned}$$

Set first derivative of  $x$  to zero:

$$\begin{aligned} 0 + \nabla f(x^{(k)}) + \frac{1}{\eta} (x - x^{(k)}) &= 0 \\ x^{(k+1)} &= x^{(k)} - \eta \nabla f(x^{(k)}) \end{aligned}$$

# Gradient ascent method

- Greedy algorithm of the steepest ascent (gradient method) for solving problems with several variables
- Function must be continuously differentiable

Input: concave, continuously differentiable function  $f: \mathbb{R}^n \rightarrow \mathbb{R}$ ,  
feasible start point  $x^{(1)} \in \mathbb{R}^n$  and parameter  $\varepsilon > 0$

```
k = 1
While( $\|\nabla f(x^{(k)})\| \geq \varepsilon$ ) {
    ➤ Compute  $\eta^* > 0$  such that  $f(x^{(k)} + \eta \nabla f(x^{(k)}))$  is maximal
    ➤ Set  $x^{(k+1)} = x^{(k)} + \eta^* \nabla f(x^{(k)})$ 
    ➤ k ++
}
```

# Gradient ascent method

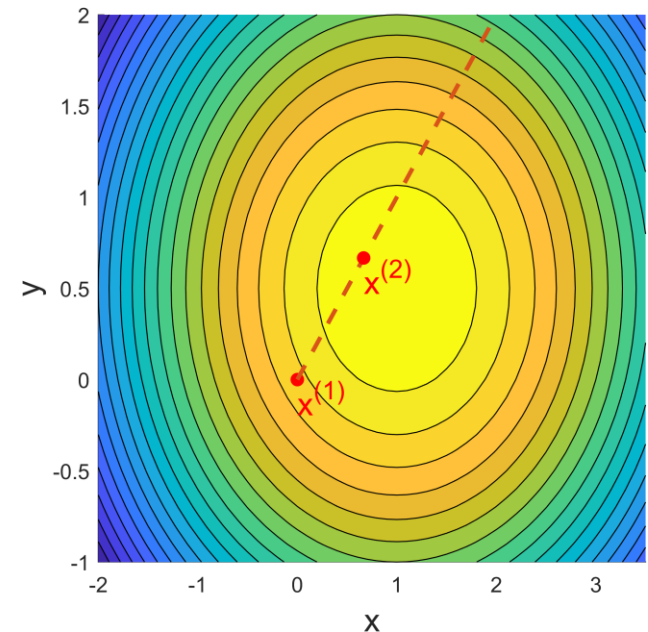
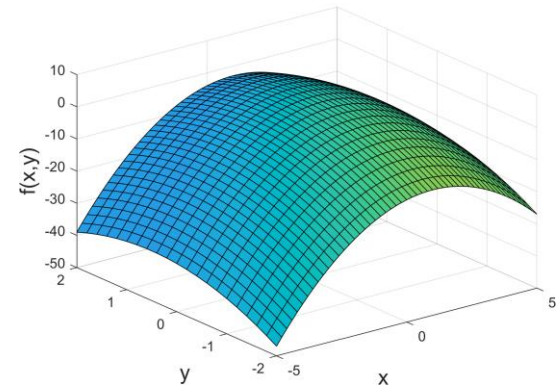
$$f(x_1, x_2): 2(x_1 + x_2) - x_1^2 - 2x_2^2$$

$$\nabla f(x) = \begin{pmatrix} 2(1 - x_1) \\ 2(1 - 2x_2) \end{pmatrix}$$

- Start at  $x^{(1)} = (0,0)$ ,  $f(x^{(1)}) = 0$ ,  $\nabla f(x^{(1)}) = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$
  - Maximize  $f(0 + 2\eta, 0 + 2\eta) = 8\eta - 12\eta^2$ 
    - Step  $\eta$  in the direction (line) of the gradient
  - Set derivative to 0:  $8 - 24\eta = 0$
- maximum point is at  $\frac{1}{3}$

$$\rightarrow x^{(2)} = \left(\frac{2}{3}, \frac{2}{3}\right), f(x^{(2)}) = \frac{4}{3}, \nabla f(x^{(2)}) = \begin{pmatrix} \frac{2}{3} \\ -\frac{2}{3} \end{pmatrix}$$

...



# Today you will learn...

- **Newton's method** as an optimization algorithm.
- the basics of **constrained convex optimization**.
- the idea of **Lagrangian duality** and the **KKT conditions**.
- how to solve constrained problems using the **projected gradient descent method**.



# Second order methods

The idea behind the **gradient method** is based on the **first order** Taylor approximation to approximate a function locally.

This approximation does not consider the curvature of the function or the rate of change of the gradient. Therefore, it is possible to overshoot the optimum using a constant step size.

Instead, **second order methods** use the first three terms of the multidimensional Taylor approximation.

$$f(x) \approx f(x^{(0)}) + \nabla f(x^{(0)})^T (x - x^{(0)}) + \frac{1}{2!} (x - x^{(0)})^T H_f(x^{(0)}) (x - x^{(0)})$$

$$H_f(x^{(0)}) = \nabla^2 f(x^{(0)}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(x^{(0)}) & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(x^{(0)}) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(x^{(0)}) & \dots & \frac{\partial^2 f}{\partial x_n^2}(x^{(0)}) \end{bmatrix}$$

# Second order methods

The idea of Newton's method is now to minimize this second order approximation. This is reasonable if  $f(x)$  is convex.

## Newton's method

$$f(x) \approx f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)}) + \frac{1}{2} f''(x^{(k)}) (x - x^{(k)})^2$$

- Setting the derivative of this approx. to zero:  $0 = f'(x^{(k)}) + f''(x^{(k)})(x - x^{(k)})$
- Solve for  $x$ :  $x^{(k+1)} = x^{(k)} - f'(x^{(k)})/f''(x^{(k)})$
- Multidimensional update rule:  $x^{(k+1)} = x^{(k)} - \left(H_f(x^{(k)})\right)^{-1} \nabla f(x^{(k)})$

For comparison, the update rule of the gradient descent method:

$$x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$$

Newton's method for optimization can also be understood as an algorithm that searches for zeros of the first derivative:  $f'(x) = 0$

# Minimal example for the Newton's method

Objective function:

$$\min f(x_1, x_2) = 3 + (x_1 - 1.5x_2)^2 + (x_2 - 2)^2$$

Start point:  $x^{(0)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Newton step:  $\Delta x = -[H(x_0)]^{-1} \nabla f(x_0)$

Compute gradient:

$$\nabla f(x^{(0)}) = \begin{pmatrix} 2(x_1 - 1.5x_2) \\ -3(x_1 - 1.5x_2) + 2(x_2 - 2) \end{pmatrix}$$

$$\nabla f(x^{(0)}) = \begin{pmatrix} -1 \\ -0.5 \end{pmatrix}$$

Compute the inverse of the Hessian matrix:

$$H(x^{(0)}) = \begin{bmatrix} 2 & -3 \\ -3 & 6.5 \end{bmatrix}$$

$$H^{-1}(x^{(0)}) = \begin{bmatrix} 1.625 & 0.75 \\ 0.75 & 0.5 \end{bmatrix}$$

Newton step:

$$\Delta x = - \begin{bmatrix} 1.625 & 0.75 \\ 0.75 & 0.5 \end{bmatrix} \begin{pmatrix} -1 \\ -0.5 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

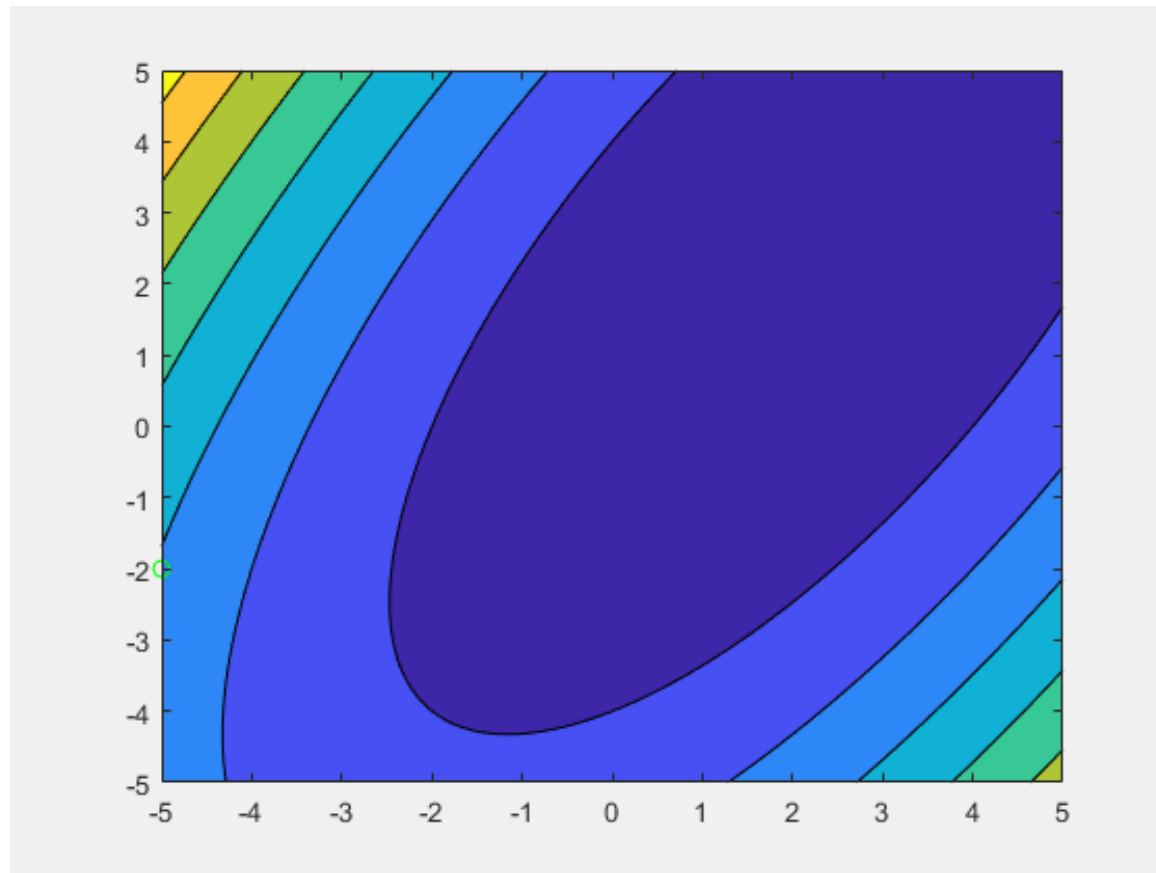
Next iteration:

$$x^{(1)} = x^{(0)} + \Delta x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

until a termination condition is fulfilled.

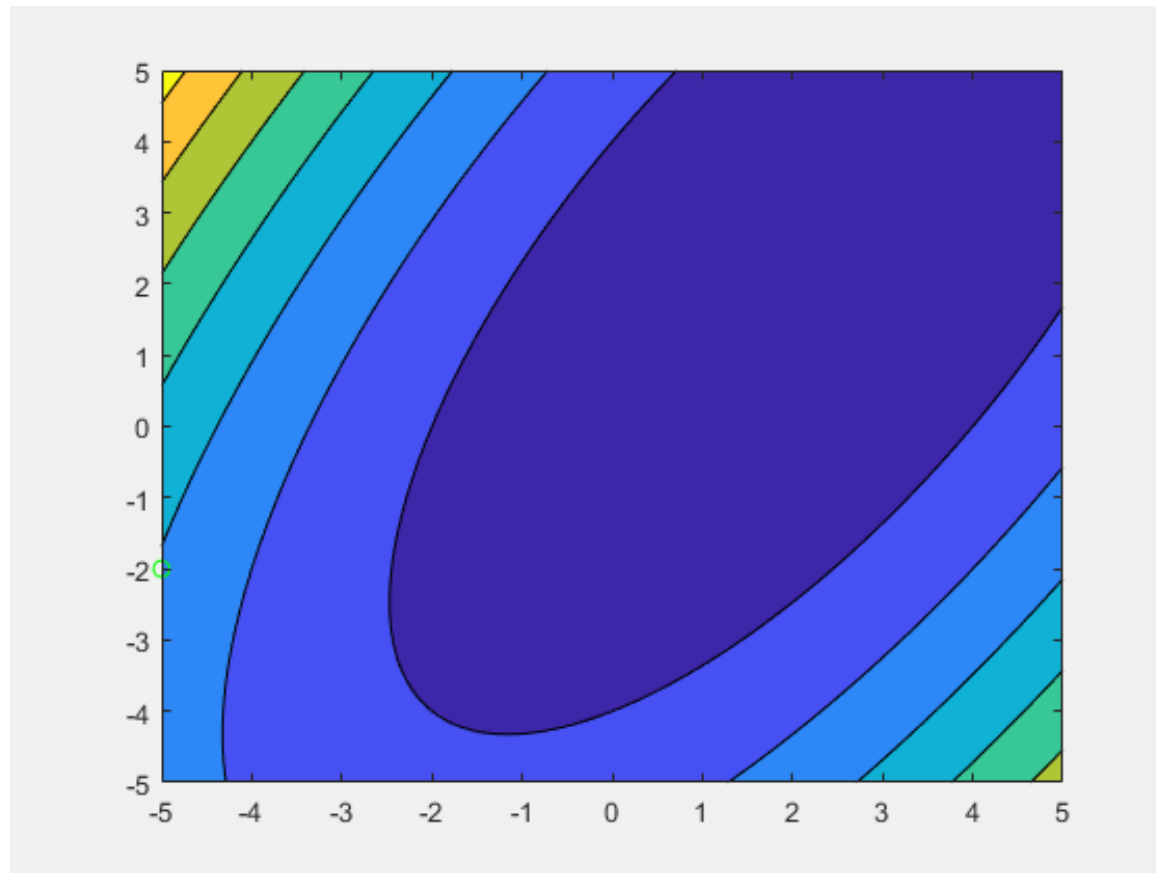
- The Newton method is very effective but requires a  $C^2$ -function.
- If the objective function is not convex, the Hessian matrix does not point in the direction of the steepest ascent.
- Calculating the inverse of the Hessian matrix is computationally expensive.

# Newton's method



$$f(x_1, x_2) = 0.1x_1^4 + (x_1 - 2)^2 - (x_1 - 2)(x_2 - 2) + 0.5(x_2 - 2)^2$$

# Gradient method



$$f(x_1, x_2) = 0.1x_1^4 + (x_1 - 2)^2 - (x_1 - 2)(x_2 - 2) + 0.5(x_2 - 2)^2$$

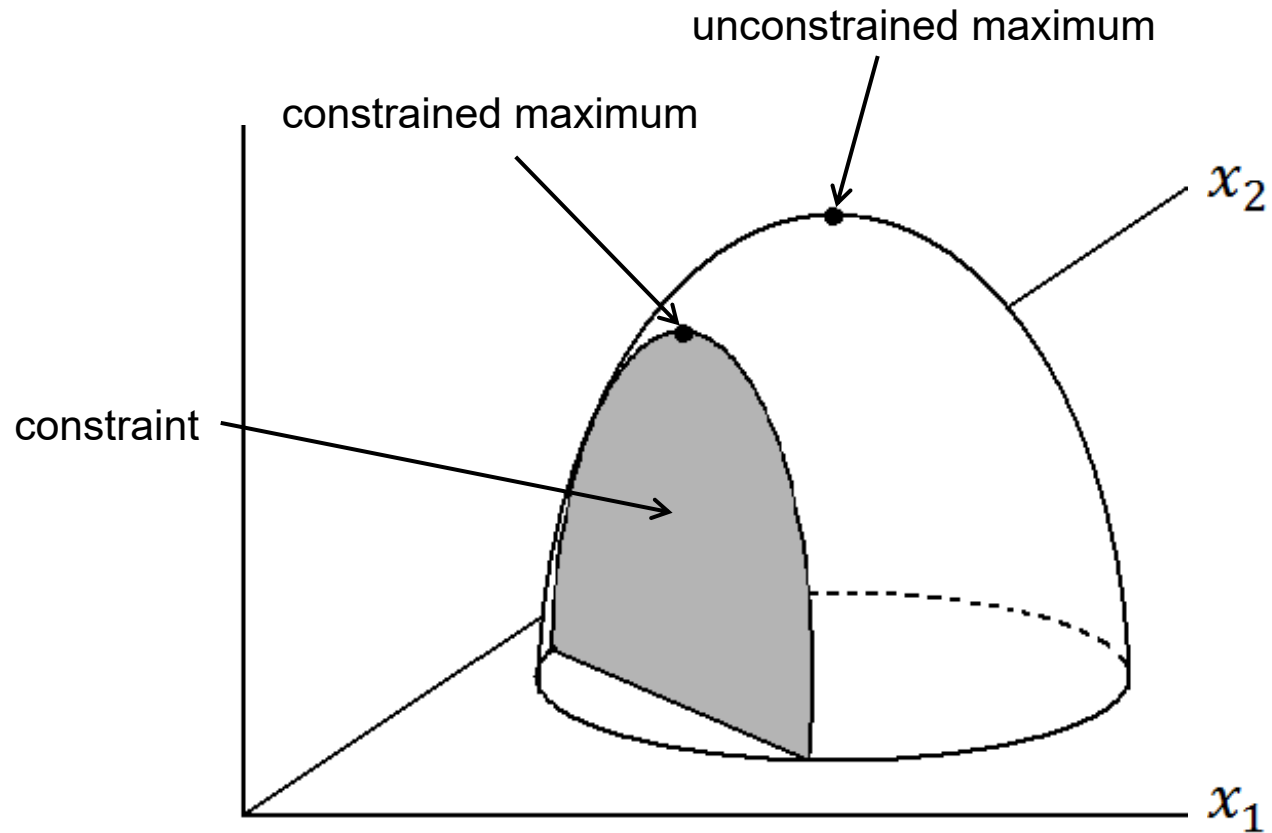
with decreasing step size

# Multivariate Newton method

Update rule:  $x^{(k+1)} = x^{(k)} - \left(H_f(x^{(k)})\right)^{-1} \nabla f(x^{(k)})$

- The inverse does not always exist and its computation is expensive ( $O(n^3)$  operations for a  $n \times n$  matrix).
- Overall, the Newton method requires more operations and more memory than the gradient method in each iteration. However, it requires fewer iterations.
- The Hessian matrix can be approximated, resulting in **Quasi-Newton** methods:
  - Davidon-Fletcher-Powell (DFP)
  - Broyden-Fletcher-Goldfarb-Shanno (BFGS)
  - Limited-memory BFGS (L-BFGS)

# Constrained convex optimization



# Convex optimization with equality constraints

The constrained optimum of a concave maximization function  $f: X \rightarrow \mathbb{R}$  with  $X \subseteq \mathbb{R}^n$  and affine functions  $g_i(x)$  can be found using **Lagrange multipliers**  $\lambda \in \mathbb{R}^m$ .

$$\begin{array}{ll} \max & f(x) \\ \text{s.t.} & g_i(x) = b_i \quad i = 1, \dots, m \\ & x \in \mathbb{R}^n \end{array}$$

Lagrangian function:  $L(x, \lambda) = f(x) - \lambda^T (g(x) - b)$

$$L(x, \lambda) = f(x) - \sum_{i=1}^m \lambda_i (g_i(x) - b_i) = f(x) - \lambda_1 (g_1(x) - b_1) - \dots - \lambda_m (g_m(x) - b_m)$$

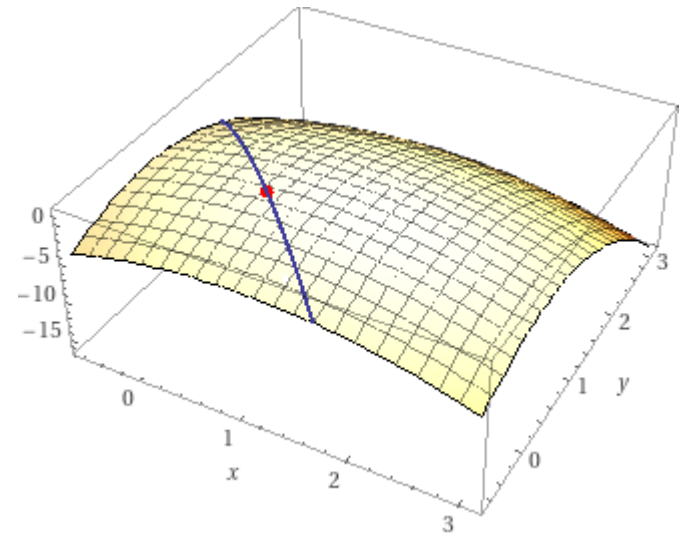
For every point  $x \in X \subseteq \mathbb{R}^n$  that maximizes  $f$ , there are Lagrange multipliers that satisfy the following conditions at critical points:

$$\frac{\partial L}{\partial x_1} = \frac{\partial L}{\partial x_2} = \dots = \frac{\partial L}{\partial x_n} = \frac{\partial L}{\partial \lambda_1} = \frac{\partial L}{\partial \lambda_2} = \dots = \frac{\partial L}{\partial \lambda_m} = 0$$

If the conditions are satisfied, then  $x$  maximizes a concave function  $f$  over  $X \subseteq \mathbb{R}^n$ .

# Lagrange multipliers

Example:  $\max \quad 2(x_1 + x_2) - x_1^2 - 2x_2^2$   
 s.t.  $x_1 + x_2 = 1$



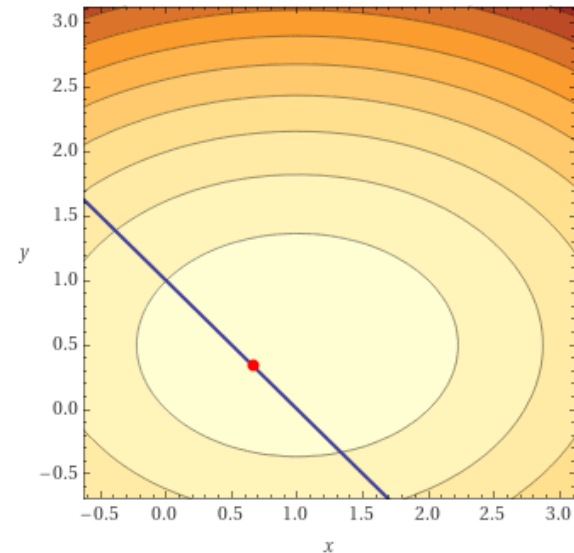
Lagrangian function:

$$L(x, \lambda) = 2(x_1 + x_2) - x_1^2 - 2x_2^2 - \lambda(x_1 + x_2 - 1)$$

Optimization via 1st order condition:  $\nabla L = 0$

Observation:  $\lambda(x_1 + x_2 - 1)$  can be subtracted or added under equality constraints.

The consequence is only a sign change in  $\lambda$ .



# Lagrange multipliers

## Theorem:

Let  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  be a concave differentiable function and each  $g_i(x)$  be an affine-linear function. Each point that satisfies

$$\frac{\partial L}{\partial x_1} = \frac{\partial L}{\partial x_2} = \dots \frac{\partial L}{\partial x_n} = \frac{\partial L}{\partial \lambda_1} = \frac{\partial L}{\partial \lambda_2} = \dots \frac{\partial L}{\partial \lambda_m} = 0$$

is an optimal solution to the maximization problem.

Computing the saddle point (maximum):

$$L(x, \lambda) = 2(x_1 + x_2) - x_1^2 - 2x_2^2 - \lambda(x_1 + x_2 - 1)$$

$$\frac{\delta L}{\delta x_1} = 2 - 2x_1 - \lambda = 0$$

$$\frac{\delta L}{\delta x_2} = 2 - 4x_2 - \lambda = 0$$

$$\frac{\delta L}{\delta \lambda} = -x_1 - x_2 + 1 = 0$$

3 unknowns:  $x_1, x_2, \lambda$   
3 equalities

# Solution

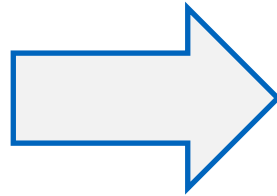
$$\frac{\delta L}{\delta x_1} = 2 - 2x_1 - \lambda = 0$$

$$\frac{\partial L}{\partial x_2} = 2 - 4x_2 - \lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = -x_1 - x_2 + 1 = 0$$

$$2 - 2x_1 - \lambda = 0$$

$$x_1 = \frac{1}{2}(2 - \lambda)$$



$$2 - 4x_2 - \lambda = 0$$

$$x_2 = \frac{1}{4}(2 - \lambda)$$

$$x_1 + x_2 = 1$$

$$x_1 = \frac{2}{3}, x_2 = \frac{1}{3}, \lambda = \frac{2}{3}, z = \frac{4}{3}$$

$$\frac{1}{2}(2 - \lambda) + \frac{1}{4}(2 - \lambda) = 1$$

$$\lambda = \frac{2}{3}$$

$$x_1 = \frac{2}{3}, x_2 = \frac{1}{3}$$

# Hessian matrix

$$H(x_1, x_2) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & -4 \end{pmatrix}$$

$$\begin{array}{ll} \max & 2(x_1 + x_2) - x_1^2 - 2x_2^2 \\ \text{s.t.} & x_1 + x_2 = 1 \\ & x_1, x_2 \geq 0 \end{array}$$

$H(x)$  is negative definite  $\Rightarrow$  maximum

- $L$  has a maximum at  $x_1 = \frac{2}{3}, x_2 = \frac{1}{3}$  when  $\lambda = \frac{2}{3}$  is fixed.
- $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$  is a saddle point (see the following slides)
- $\left(\frac{2}{3}, \frac{1}{3}\right)$  is the maximum point of  $f$ , with maximum  $\frac{4}{3}$
- The variable  $\lambda = \frac{2}{3}$  shows how changing the constraint  $x_1 + x_2 = 1$  by a small amount affects the objective function value. For equality constraints,  $\lambda$  can take positive and negative values, since any deviation is penalized.

# Geometric interpretation

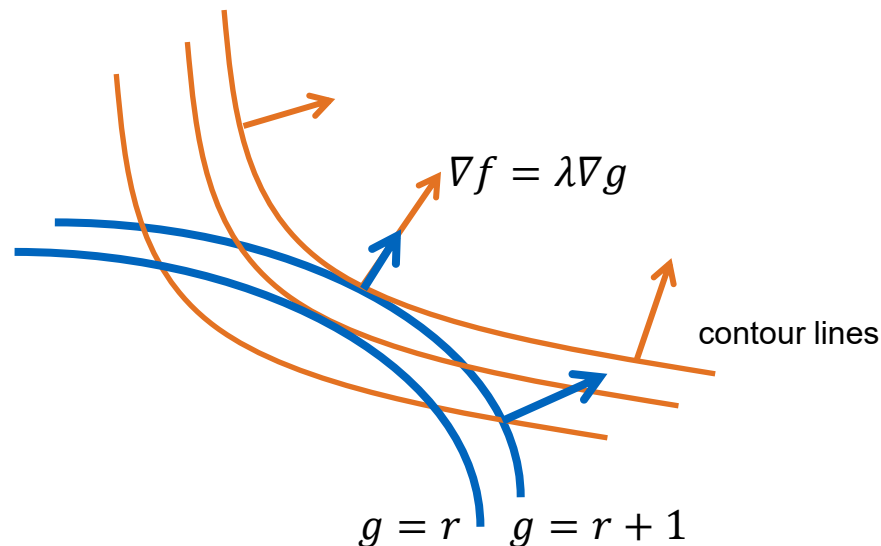
$$L = f(x) - \lambda^T (g(x) - b_i)$$

Note that component-wise:  $\frac{\partial L}{\partial x_j} = 0 = \frac{\partial f}{\partial x_j} - \sum_{i=1}^m \lambda_i \frac{\partial g_i}{\partial x_j}$

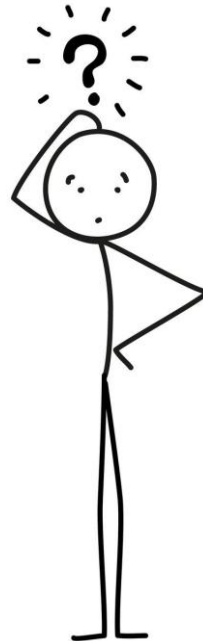
$\frac{\partial L}{\partial x_j} = 0 \forall j = 1, \dots, n$  is therefore equivalent to  $\nabla f = \sum_{i=1}^m \lambda_i \nabla g_i$ .

In other words,  $\nabla f$  is a linear combination of the gradients of the constraints.

With only one constraint, it can be seen that the gradient of the objective function equals that of the constraint.

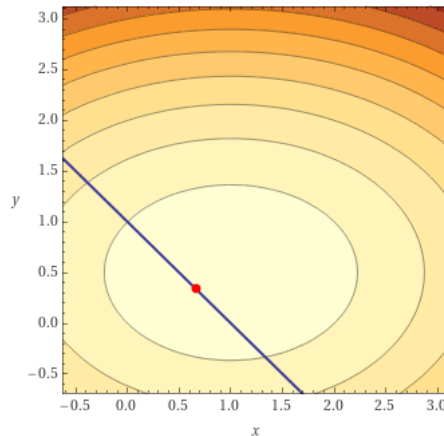


$$\begin{aligned} \max \quad & 2(x_1 + x_2) - x_1^2 - 2x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 = 1.1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

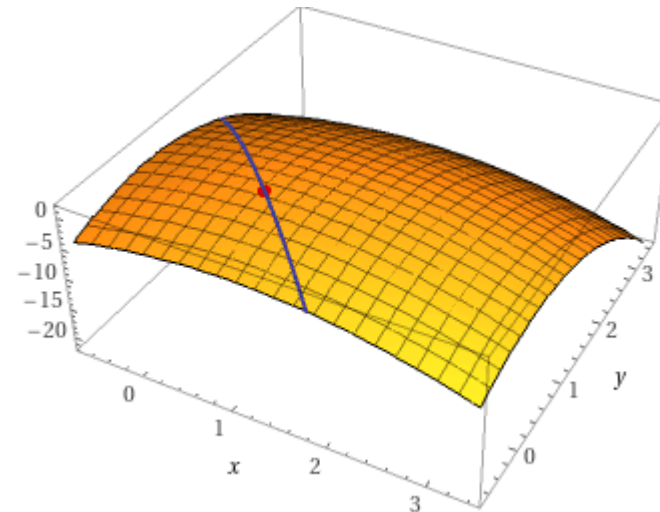


# Marginal changes of the constraint: sensitivity

$$\begin{aligned} \max \quad & 2(x_1 + x_2) - x_1^2 - 2x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 = 1.1 \\ & x_1, x_2 \geq 0 \end{aligned}$$



$$\begin{aligned} x_1 + x_2 &= 1 \\ x_1 &= \frac{2}{3}, x_2 = \frac{1}{3}, \lambda = \frac{2}{3}, z = \frac{4}{3} \end{aligned}$$



$$\begin{aligned} x_1 + x_2 &= 1.1 = 1 + \Delta \\ x_1 &= 0.733, x_2 = 0.367, \lambda = 0.533, z' = 1.39333 \\ \text{Previous } z &= 4/3 = 1.33333 \\ \text{Difference } z' - z &= 0.06 \approx \lambda \Delta = \frac{2}{3} \cdot 0.1 = 0.0667 \end{aligned}$$

For a small change in the right-hand side, the change in the optimal objective is approximately lambda times that change.

# Slater's conditions for *concave* optimization

$$\begin{array}{ll} \max & f(x) \\ \text{s.t.} & g_i(x) = 0 \quad i = 1, \dots, l \\ & h_j(x) \leq 0 \quad j = 1, \dots, m \end{array}$$

Slater's conditions represent the most important example of **regularity conditions**. They allow for **strong duality** and require that:

- $f(x)$  is concave
- Non-affine  $h_j(x)$  are convex
- $g_i(x)$  is affine
- There exists a feasible point  $x \in X$  satisfying all equality constraints and satisfying all non-affine inequality  $h_j(x)$  constraints strictly

=> Then all points  $x \in X$  are regular.

# Lagrangian duality with equality constraints

**Primal (P):**

$$\begin{aligned} \max \quad & f(x) \\ \text{s.t.} \quad & g_i(x) = b_i \quad i = 1, \dots, m \\ & x \in \mathbb{R}^n \end{aligned}$$

Primal (concave) function

$$\begin{aligned} p(x) &= \inf_{\lambda} L(x, \lambda) = f(x) - \sum_{i=1}^m \lambda_i (g_i(x) - b_i) \\ &= \begin{cases} f(x) & \text{if } g_i(x) = b_i \quad \forall i = 1, \dots, m \\ -\infty & \text{otherwise} \end{cases} \end{aligned}$$

We define the **dual problem (D)** as the following convex function:

$$\min \quad d(\lambda) \quad (\text{The dual has one variable per constraint})$$

$$d(\lambda) = \sup_x L(x, \lambda) = f(x^*) - \sum_{i=1}^m \lambda_i (g_i(x^*) - b_i)$$

We can solve the primal problem by finding an optimal point  $\lambda^*$  of the dual problem and then computing  $x^*$  as the solution of  $\max p(x)$ . The solution is simultaneous.

# Example

Primal problem:  $\max -(x_1^2 + x_2^2), \text{ s.t. } x_1 + x_2 - 1 = 0$  (casts  $\min x_1^2 + x_2^2$  as a maximization)

Lagrangian:  $L(x_1, x_2, \lambda) = -(x_1^2 + x_2^2) + \lambda(x_1 + x_2 - 1)$

Dual function:  $d(\lambda) = \sup_{x_1, x_2} L(x_1, x_2, \lambda)$

Finding the supremum leads to  $\frac{\partial L}{\partial x_1} = -2x_1 + \lambda = 0 \Rightarrow x_1 = \frac{\lambda}{2}$

and  $\frac{\partial L}{\partial x_2} = -2x_2 + \lambda = 0 \Rightarrow x_2 = \frac{\lambda}{2}$

Substituting back into  $x_1 + x_2 - 1 = 0$

$$\frac{\lambda}{2} + \frac{\lambda}{2} = 1 \Rightarrow \lambda = 1, x_1 = \frac{1}{2}, x_2 = \frac{1}{2}$$

Dual function value

$$L(x_1, x_2, \lambda) = -\left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right) + 1\left(\frac{1}{2} + \frac{1}{2} - 1\right) = -\frac{1}{2}$$

Dual problem: minimize  $d(\lambda) = \inf_{\lambda} \sup_{x_1, x_2} L(x_1, x_2, \lambda) = \min d(1) = -\frac{1}{2}$

=> Assuming a different  $\lambda$  would lead to a violation of the constraint or a suboptimal solution.

# Lagrangian duality

Primal problem (maximization)

$$p(x) = \inf_{\lambda} L(x, \lambda) = \inf_{\lambda} (f(x) - \sum_{i=0} \lambda_i (g_i(x) - b_i))$$

$$d(\lambda) = \sup_x (f(x) - \sum_{i=0} \lambda_i (g_i(x) - b_i))$$

The optimal value of the primal problem:

$$p^* = \sup_x \inf_{\lambda} L(x, \lambda) = \sup_x p(x)$$

The optimal value of the Lagrangian dual function:

$$d^* = \inf_{\lambda} \sup_x L(x, \lambda) = \inf_{\lambda} d(\lambda)$$

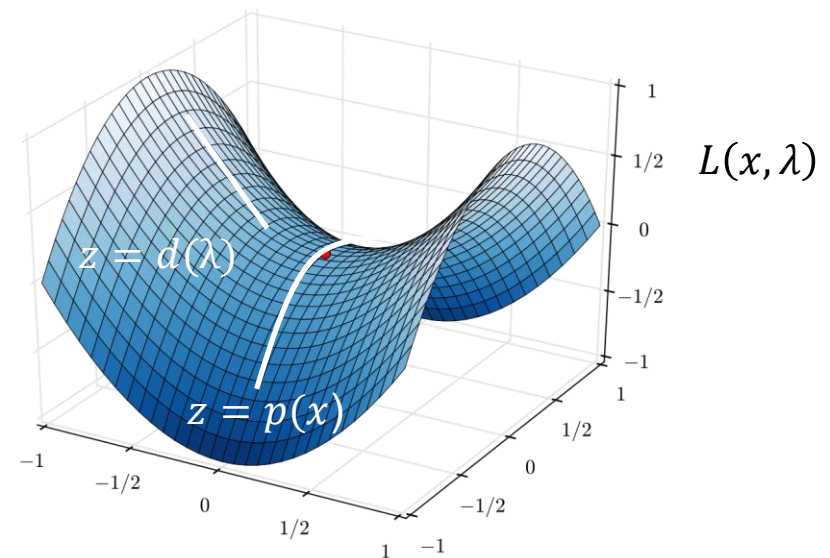
**Weak duality:** Dual as upper bound for the primal maximization problem:

$$d^* = \inf_{\lambda} \sup_x L(x, \lambda) \geq \sup_x \inf_{\lambda} L(x, \lambda) = p^*$$

**Strong duality:** applies if the Slater's conditions hold:

$$\inf_{\lambda} \sup_x L(x, \lambda) = \sup_x \inf_{\lambda} L(x, \lambda)$$

Saddle point of the Lagrangian function  
=> strong duality



# Karush-Kuhn-Tucker conditions for convex optimization with inequalities

$$\begin{array}{ll} \max & f(x) \\ \text{s.t.} & g_i(x) = 0 \quad i = 1, \dots, l \\ & h_j(x) \leq 0 \quad j = 1, \dots, m \end{array}$$

**Karush-Kuhn-Tucker (KKT) conditions** generalize the method of Lagrangian multipliers to inequalities.

- $f(x)$  is concave/convex.
- If  $g_i(x)$  and  $h_j(x)$  are affine-linear and  $x'$  is a maximum  $\Rightarrow$  KKT hold.
- If the **Slater's conditions** hold and  $x'$  is a maximum  $\Leftrightarrow$  KKT hold.

When Slater's conditions hold, all points in  $X$  are regular and the KKT conditions imply a **global maximum**. **Strong duality** holds.

Observation: If  $f$  is concave,  $-f$  is convex.

# Karush-Kuhn-Tucker (KKT) conditions

$$\begin{aligned}
 \max \quad & f(x) \\
 \text{s.t.} \quad & g_i(x) = 0 \quad i = 1, \dots, m \\
 & h_j(x) \leq 0 \quad j = 1, \dots, p \\
 & x \in \mathbb{R}^n
 \end{aligned}$$

Lagrangian function  $L = f(x) - \lambda^T g(x) - \mu^T h(x)$ . KKT conditions for optimal point  $x'$ :

$$\nabla_x f(x') - \sum_i \lambda_i \nabla_x g_i(x') - \sum_j \mu_j \nabla_x h_j(x') = 0$$

$$g_i(x') = 0$$

$$h_j(x') \leq 0$$

$$\mu_j \geq 0$$

$$\mu_j h_j(x') = 0$$

1. The gradient of the Lagrangian function should be 0 in opt.
2. Primal feasibility for all  $i$
3. Primal feasibility for all  $j$
4. The Lagrangian multipliers of inequalities  $\geq 0$  shall not be negative (as opposed to  $\lambda_i$ ), aka. dual feasibility
5. Complementarity slackness for inequalities for all  $j$

KKT constraints cannot always be solved analytically.

Many optimization algorithms try to find the KKT conditions numerically.

# KKT conditions for our example with $\leq$

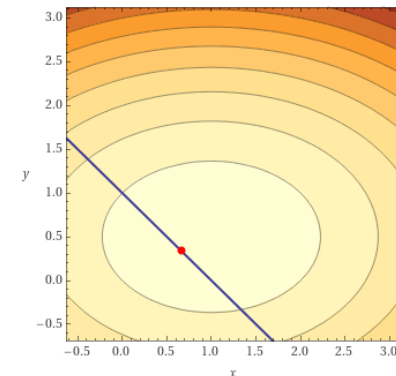
$$\begin{aligned} \max \quad & 2(x_1+x_2) - x_1^2 - 2x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 - 1 \leq 0 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$L = 2(x_1+x_2) - x_1^2 - 2x_2^2 - \mu(x_1 + x_2 - 1)$   
 Subtracting  $\mu(x_1 + x_2 - 1)$  such that  
 a violation of the constraint is penalized.

1.  $L_{x_1} = 2 - 2x_1' - \mu' = 0$   
 $L_{x_2} = 2 - 4x_2' - \mu' = 0$
2. no equality constraint
3.  $x_1 + x_2 - 1 \leq 0$
4.  $\mu' \geq 0$
5.  $\mu'(x_1' + x_2' - 1) = 0$

1. The gradient of the Lagrangian function should be 0 in opt.
2. Primal feasibility for all  $i$
3. Primal feasibility for all  $j$
4. The Lagrangian multipliers of inequalities  $\geq 0$  may not be negative (as opposed to  $\lambda_i$ )
5. Complementarity for inequalities for all  $j$

The point  $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$  satisfies the KKT conditions.



# Another example

$$\min x^3 \text{ s.t. } x \geq 0$$

Lagrange function

$$L(x, \mu) = x^3 + \mu(-x) = x^3 - \mu x$$

Stationarity

$$L_x = 3x^2 - \mu = 0$$

Primal feasibility

$$x \geq 0$$

Dual feasibility

$$\mu \geq 0$$

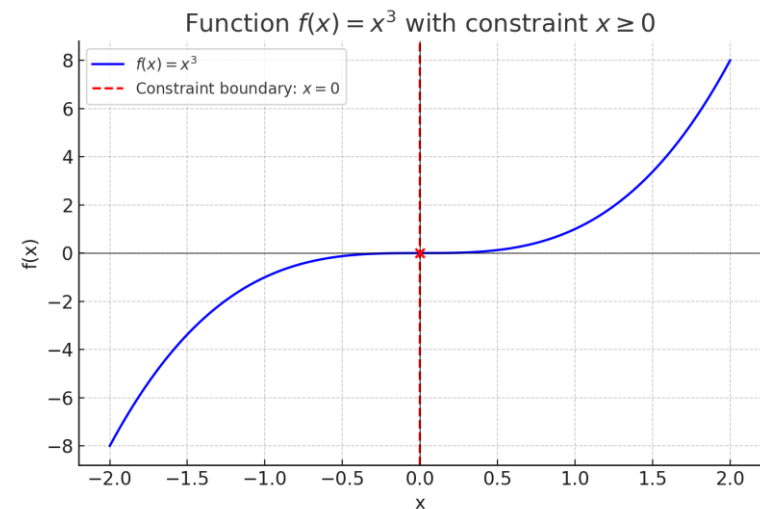
Complementary slackness (CS)

$$\mu x = 0$$

From CS either  $x = 0$  or  $\mu = 0$

If  $x = 0$ , then  $3x^2 - \mu = 0 \Rightarrow \mu = 0$ : this is a KKT point and a minimizer of the constrained problem.

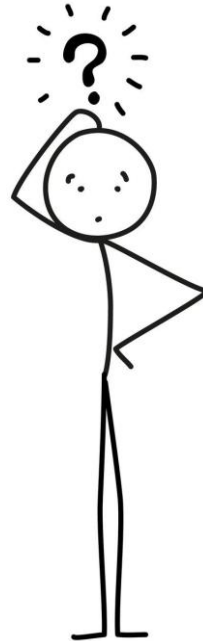
The function is **not convex**, and KKT are necessary but not always sufficient in these settings!



Maximize  $f(x_1, x_2) = -x_1^2 - x_2^2 + 14x_1 + 6x_2$

s.t.  $h_1(x_1, x_2) = x_1 + x_2 - 2 \leq 0$

$h_2(x_1, x_2) = x_1 + 2x_2 - 3 \leq 0$



**Maximize**  $f(x_1, x_2) = -x_1^2 - x_2^2 + 14x_1 + 6x_2$

**s.t.**  $h_1(x_1, x_2) = x_1 + x_2 - 2 \leq 0$

$h_2(x_1, x_2) = x_1 + 2x_2 - 3 \leq 0$

$$L(x_1, x_2, \mu_1, \mu_2) = -x_1^2 - x_2^2 + 14x_1 + 6x_2 - \mu_1(x_1 + x_2 - 2) - \mu_2(x_1 + 2x_2 - 3)$$

$$L_{x_1} = -2x_1 + 14 - \mu_1 - \mu_2 = 0$$

$$L_{x_2} = -2x_2 + 6 - \mu_1 - 2\mu_2 = 0$$

**Case 1** (both constraints are binding):

$$x_1 + x_2 - 2 = 0$$

$$x_1 + 2x_2 - 3 = 0$$

Solution of the system of equations:

$$x_1 = 1, x_2 = 1, \mu_1 = 20, \mu_2 = -8 \text{ (violates the KKT constraints)}$$

**Case 2** (none of the constraints is binding):

$$L_{x_1} = -2x_1 + 14 = 0 \Rightarrow x_1 = 7$$

$$L_{x_2} = -2x_2 + 6 = 0 \Rightarrow x_2 = 3$$

With these values, the constraints would be violated.

**Case 3** (only the first constraint is binding):  $x_1 = 3, x_2 = -1, \mu_1 = 8$

$$h_1(3, -1) = 0, h_2(3, -1) = -2, f(3, -1) = 26 \text{ (optimal)}$$

$$\begin{aligned} -2x_1 + 14 - \mu_1 &= 0 \\ -2x_2 + 6 - \mu_1 &= 0 \\ x_1 + x_2 - 2 &= 0 \end{aligned}$$

**Case 4** If only the constraint  $h_2$  was active, the solution would again not be feasible.

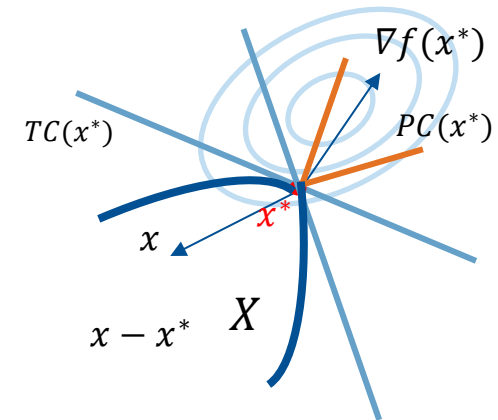
# Other optimality conditions

Tangential cone (TC):

$$TC_X := \{z \in \mathbb{R}^d : z = \alpha(x - x^*), x \in X, \alpha \geq 0\}$$

Normal cone or polar cone (PC):

$$PC_X := \{y \in \mathbb{R}^d : \langle y, z \rangle \leq 0, \forall z \in TC_X(x^*)\}$$



Let  $X \subseteq \mathbb{R}^n$  be a nonempty, closed and convex set and  $f: X \rightarrow \mathbb{R}$ .

Necessary optimality conditions for **maximization** under constraints are

$$\nabla f(x^*)^T (x - x^*) \leq 0 \quad \forall x \in X$$

... and for **minimization** under constraints are:

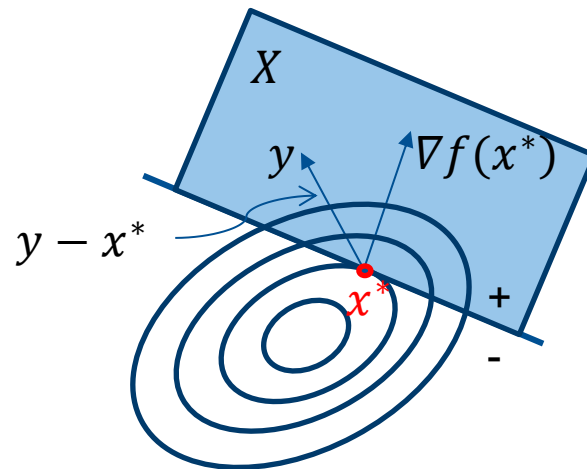
$$\nabla f(x^*)^T (x - x^*) \geq 0 \quad \forall x \in X$$

Alternativ: Compute  $x \in X$  s.t.  $-\nabla f(x) \in PC(x)$  (polar cone).

# First-order optimality in convex optimization

**Theorem (Min):** Let  $X \subseteq \mathbb{R}^n$  be a non-empty, closed and convex set and  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  be continuously differentiable and (pseudo-)convex. Then: If  $x^* \in X$  is a minimum of the above optimization problem,  $\nabla f(x^*)^T (x - x^*) \geq 0 \quad \forall x \in X$ .

This is a minimal necessary condition (while KKT gives a complete set of conditions)



# Can we always solve the KKT analytically?

Solving the KKT conditions analytically, as in the example, often leads to infeasible or overly complex systems.

- For  $m$  constraints, we might need to check  $2^m$  combinations of constraints!
- Solving stationarity conditions can be costly except for simple (quadratic) functions.
- Degeneracy: Sometimes several constraints meet at the optimum, some multipliers are zero even though the constraint is active, or the solution lies at a corner where the active set is not obvious.

Various numerical optimization methods were devised.

- Gradient-based methods, Interior-point methods, Frank-Wolfe method, augmented Lagrangian methods, etc.

Let's focus on convex objective functions with simple linear constraints, on which we can simply project onto.

We look at gradient-based methods, in particular, projected gradient descent and entropic mirror descent.

# Gradient *descent* method with constraints

## Pseudocode:

$k = 1$

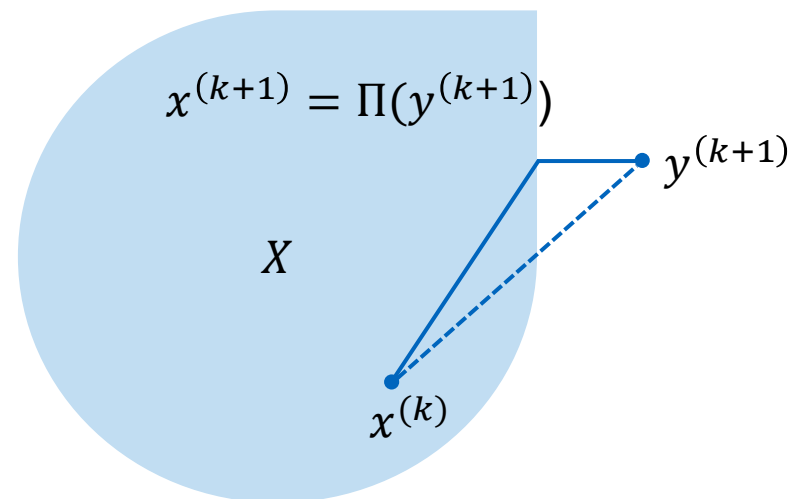
While( $\|\nabla f(x^{(k)})\| \geq \varepsilon$ ) {

- $y^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$

- $x^{(k+1)} = \Pi(y^{(k+1)}) = \arg \min_{x \in X} \|y^{(k+1)} - x\|$  //Euclidean projection

- $k++$

}

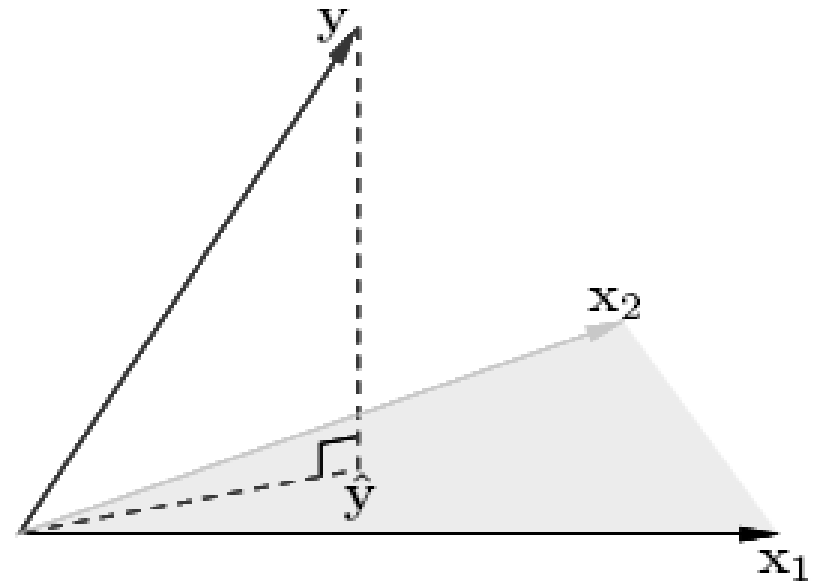


# Projection

**Definition (Projection):**

The set  $X \subset \mathbb{R}^n$  is non-empty, closed and convex. For a fixed  $y \in \mathbb{R}^n$  we search a point  $\hat{y} \in X$ , with the smallest distance to  $y$  (wrt. the Euclidean norm), i.e. we solve the minimization problem

$$\Pi_X(y) = \min_{\hat{y} \in X} \|y - \hat{y}\|^2$$



Source: Hastie et al. 2016, p. 46

Vector  $y - \hat{y}$  is orthogonal to this subspace.

# Projection

$\mathbf{X}$  is  $n \times k$  with  $k \leq n$  whose columns form the basis for a subspace,  
 $y$  needs to be projected onto  $\mathbf{X}$

$$\Pi_{\mathbf{X}}(y) = \min_{\hat{y} \in \mathbf{X}} \|y - \hat{y}\|^2 = (y - \mathbf{X}\beta)^T (y - \mathbf{X}\beta)$$

$\beta \in \mathbb{R}^k$  is a vector of coefficients, determining  $\hat{y}$

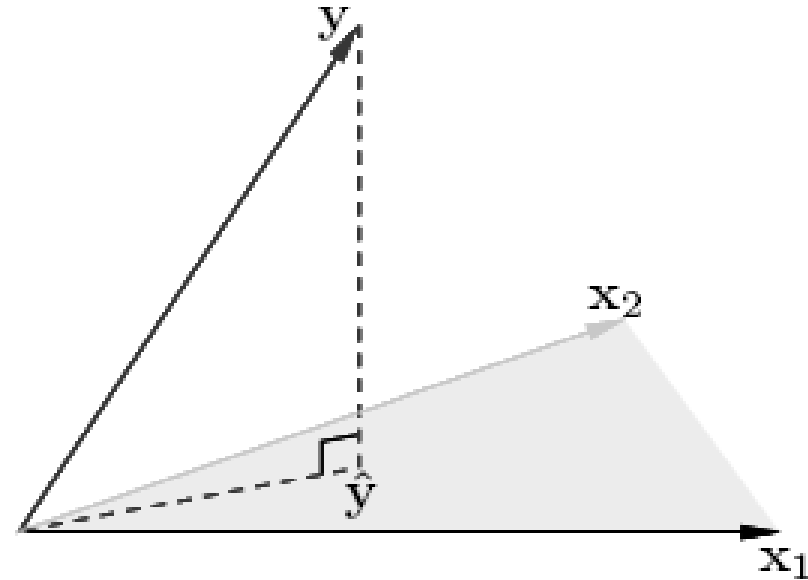
If  $\mathbf{X}$  is full rank, then  $\mathbf{X}^T \mathbf{X}$  is positive definite

$$\Pi_{\mathbf{X}}(y) = (y^T y - 2\beta^T \mathbf{X}^T y + \beta^T \mathbf{X}^T \mathbf{X} \beta)$$

$$\frac{\partial \Pi_{\mathbf{X}}(y)}{\partial \beta} = -2\mathbf{X}^T y + 2\mathbf{X}^T \mathbf{X} \beta = 0$$

$$\beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y$$

$$\hat{y} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y$$



# Example projection

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \quad y = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$$

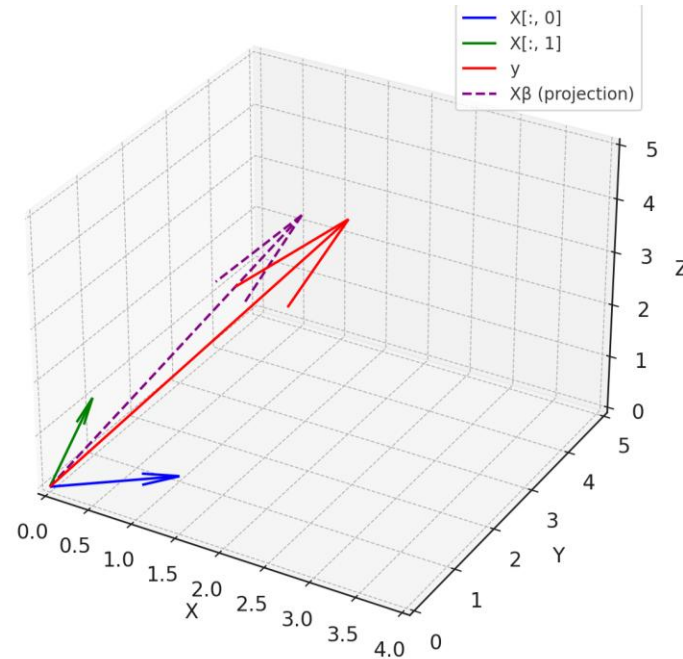
$$X^T X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\hat{y} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y$$

$$\hat{y} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$$

$$\beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$X\beta = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$$



# Example of projected gradient descent

A company can produce two products  $A$  and  $B$ .

$x$  = quantity  $A$ ,  $y$  = quantity  $B$

The maximum total (normalized) quantity that can be produced is 1:  $x + y \leq 1$

$A$  is sold for price 5 €,  $B$  for 7 €.

Costs incurred:

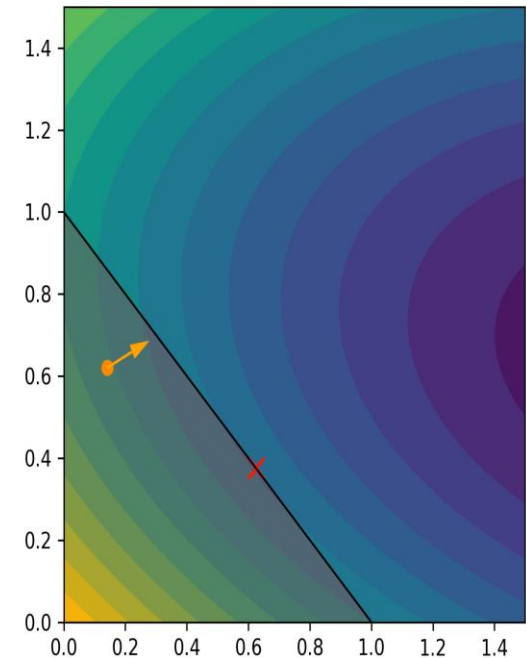
$$x^2 + xy + 4y^2 = (x, y) \begin{pmatrix} 1 & 0.5 \\ 0.5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Maximize profit ( $\Leftrightarrow$  minimize loss):

$$\min (x, y) \begin{pmatrix} 1 & 0.5 \\ 0.5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - (5x + 7y)$$

$$\text{s.t. } x + y \leq 1$$

$$x, y \geq 0$$



# Convergence of the projected gradient descent method

**Theorem:** For step size  $\eta = \frac{D}{G\sqrt{T}}$

$$\begin{aligned} y^{(k+1)} &= x^{(k)} - \eta \nabla f(x^{(k)}) \\ x^{(k+1)} &= \arg \min_{x \in X} \|y^{(k+1)} - x\| \end{aligned}$$

$$f\left(\frac{1}{T} \sum_{k \leq T} x^{(k)}\right) \leq \min_{x^* \in X} f(x^*) + \frac{DG}{\sqrt{T}}$$

Where:

$G$  is an upper bound for the norm of the gradient, i.e. steepness of the function:

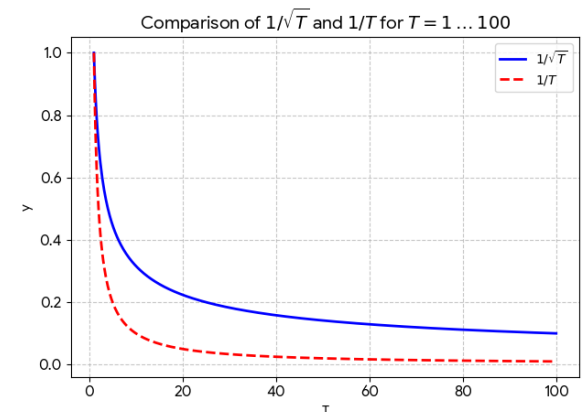
$$\|\nabla f(x^{(k)})\| \leq G$$

$D$  is the diameter of the polytope formed by the constraints:

$$\forall x, y \in X: \|x - y\| \leq D$$

In words, the function value at the average of the iterates is

at most the optimal value plus an error term. This means, the method converges at rate  $O(\frac{1}{\sqrt{T}})$ .

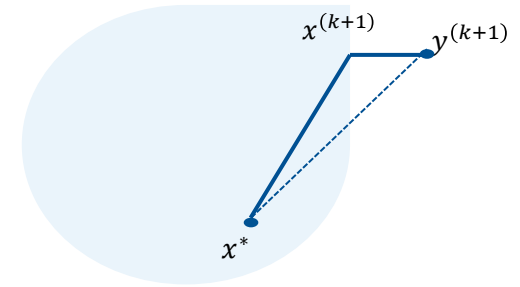


# Proof of convergence rate

Observation 1

$$\begin{aligned}\|y^{(k+1)} - x^*\|^2 &= \|x^{(k)} - \eta \nabla f(x^{(k)}) - x^*\|^2 \\ &= \|x^{(k)} - x^*\|^2 - 2\eta \nabla f(x^{(k)})(x^{(k)} - x^*) + \eta^2 \|\nabla_k\|^2\end{aligned}$$

$$(a - b)^2 = a^2 - 2ab + b^2$$



Observation 2 (Pythagoras' theorem)

$$\|x^{(k+1)} - x^*\|^2 \leq \|y^{(k+1)} - x^*\|^2$$

$$\begin{aligned}y^{(k+1)} &= x^{(k)} - \eta \nabla f(x^{(k)}) \\ x^{(k+1)} &= \arg \min_{x \in X} \|y^{(k+1)} - x\|\end{aligned}$$

Therefore:

$$\|x^* - x^{(k+1)}\|^2 \leq \|x^* - x^{(k)}\|^2 - 2\eta \nabla f(x^{(k)})(x^{(k)} - x^*) + \eta^2 \|\nabla_k\|^2$$

Use convexity:

Convexity: the average point is always above the average

This is lower-bounded by the gradient

$$f(x^*) \geq f(x^{(k)}) + \frac{1}{T} \sum_{k \leq T} \nabla f(x^{(k)})(x^{(k)} - x^*)$$

$$f\left(\frac{1}{T} \sum_{k \leq T} x^{(k)}\right) - f(x^*) \leq \frac{1}{T} \sum_{k \leq T} [f(x^{(k)}) - f(x^*)] \leq \frac{1}{T} \sum_{k \leq T} \nabla f(x^{(k)})(x^{(k)} - x^*)$$

$$\leq \frac{1}{T} \sum_{k \leq T} \frac{1}{2\eta} (\|x^* - x^{(k)}\|^2 - \|x^* - x^{(k+1)}\|^2) + \frac{\eta}{2} \|\nabla f(x^{(k)})\|^2$$

$$\|x - y\| \leq D \quad = \|x^* - x^{(1)}\|^2 - \underbrace{\|x^* - x^{(T+1)}\|^2}_{\leq 0} + \frac{\eta}{2} \|\nabla f(x^{(k)})\|^2 \quad \|\nabla f(x^{(k)})\| \leq G$$

$$\eta = \frac{D}{G\sqrt{T}}$$

$$\leq \frac{1}{T2\eta} D^2 + \frac{\eta}{2} G^2 = \frac{DG}{2\sqrt{T}} + \frac{DG}{2\sqrt{T}} \leq \frac{DG}{\sqrt{T}}$$

# Optimization has a long history

Isaac Newton (1642-1727)

- Calculus
- Optimization methods

Joseph-Louis Lagrange (1736-1813)

- Constrained optimization

Augustin-Louis Cauchy (1789-1857)

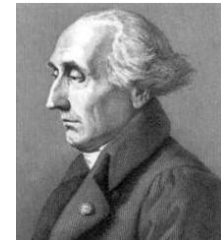
- Gradient descent for unconstrained problems

Leonhard Euler (1707-1783)

- Minimization of nonlinear functions
- Graph theory

Gottfried Leibnitz (1646-1716)

- Calculus
- Computing machine



# 20<sup>th</sup> century contributions

George Bernard Dantzig (1914-2005)

- Linear optimization and simplex (1947)



Richard Bellman (1920-1984)

- Dynamic programming



Harold William Kuhn (1925-2014) and

Albert William Tucker (1905-1995)

- Necessary conditions for optimal solutions and non-linear optimization



John von Neumann (1903-1957)

- Game theory (and much more)

